

Padamitra: Grounded Glossary Generation for Classical Sanskrit

Manoj Balaji Jagadeeshan

Sai Pragnan Marala

Pawan Goyal

Indian Institute of Technology, Kharagpur

Abstract

We introduce grounded glossary generation, a structured task requiring models to recover semantically meaningful Sanskrit phrases and produce translation-grounded meanings from a śloka - translation pair, formalizing the traditional pāṭha commentary practice as an evaluable NLP objective. We construct a benchmark of 31,316 śloka - translation glossary triples from the Vālmīki Rāmāyaṇa and Śrīmad Bhāgavatam, paired with two metrics: Jaccard for phrase recovery and Meaning Faithfulness for semantic consistency. Across zero-shot, few-shot, and instruction fine-tuned variants of Gemma-3n-E4B, Gemma-3-12B, Phi-4, and Qwen3.5-9B, instruction fine-tuning substantially outperforms prompting, while explicit segmentation yields gains. Error analysis identifies over-segmentation of sandhi and samāsa compounds as the dominant failure mode, pointing to morphological modeling as the key bottleneck for faithful Sanskrit lexical decomposition. Data¹ and Code² are available.

1 Introduction

Classical Sanskrit literature poses significant challenges for interpretation due to its morphological richness and flexible syntax. In particular, phonological fusion across word boundaries (*sandhi*) and systematic nominal compounding (*samāsa*) often produce surface forms whose segmentation into semantically meaningful units is itself a non-trivial task (Huet et al., 2009; Hellwig and Nehrdich, 2018). Classical commentators addressed this through *pāṭha* commentary - a glossary-like annotation that maps each resolved phrase in a verse (*śloka*) to its contextual meaning - a tradition that underlies modern pedagogi-

cal resources such as the IITK Valmiki Rāmāyaṇa³ and Vedabase⁴. We formalize this tradition as **grounded glossary generation**: given a Sanskrit *śloka* s and its authoritative English translation t , the task is to produce a structured glossary $\mathcal{G} = \{(k_i, v_i)\}_{i=1}^m$, where each key k_i is a semantically coherent Sanskrit phrase and each value v_i is a meaning grounded in t . Crucially, grounding in the translation distinguishes this task from unconstrained lexicographic gloss generation: the meanings must be *faithful* to the provided translation, not merely plausible in isolation. This formulation makes interpretable lexical decomposition an explicit, evaluable objective rather than an opaque by-product of end-to-end translation.

Prior Sanskrit NLP research has addressed sub-problems such as sandhi splitting, morphological analysis, and segmentation using rule-based, finite-state, and supervised methods (Huet et al., 2009; Kulkarni and Shukl, 2009; Krishna et al., 2017; Hellwig and Nehrdich, 2018). Related work has also explored Sanskrit dependency parsing (Kulkarni and Shukl, 2009), Sanskrit-English neural machine translation (Aralikatte et al., 2021), multilingual LLMs for low-resource languages (Ahuja et al., 2023), and bilingual word alignment using FastAlign (Dyer et al., 2013). However, no prior work directly addresses grounded phrase-level glossary generation as a unified task combining segmentation, semantic grounding, and lexicographic faithfulness.

Several gaps motivate this work. First, token-level alignment methods such as FastAlign cannot model Sanskrit’s fusional morphology, where a single surface token can correspond to multiple glossary keys. Second, although instruction-tuned LLMs perform well on structured prediction via in-context learning (Brown et al., 2020; Wei et al.,

¹<https://huggingface.co/collections/sanganaka/padamitra-glossary-generation>

²<https://github.com/sanganaka-iitkgp/Padamitra-Glossary-Generation>

³<https://valmiki.iitk.ac.in>

⁴<https://vedabase.io/en/library/sb/>

2022), their behavior on Sanskrit glossary generation – requiring sandhi resolution, boundary detection, and translation-grounded meaning assignment – remains unclear. Third, standard MT metrics (e.g., BLEU, chrF) are ill-suited for evaluating phrase-level key recovery and per-entry semantic faithfulness. Finally, the impact of explicit segmentation supervision versus end-to-end learning has not been established for this task.

To address these gaps, we construct a benchmark of 31,316 Sanskrit *śloka*–translation–glossary triples from the Valmiki Rāmāyaṇa and Śrīmad Bhāgavatam, and introduce two evaluation metrics: **Jaccard** for glossary key recovery and **Meaning Faithfulness** for translation-grounded semantic consistency. We benchmark zero-shot, few-shot, and instruction fine-tuned variants of gemma-3n-E4B-it, gemma-3-12b-it, phi-4(14B), Qwen3.5-9B, showing that instruction fine-tuning (IFT) substantially outperforms prompting-based approaches. Further ablations and qualitative analysis reveal that explicit segmentation provides limited gains, while over-segmentation of *sandhi* and *samāsa* compounds remains the dominant failure mode, meanwhile the IFT models are susceptible to word-order.

2 Problem Statement

Given a Sanskrit *śloka*(s) and its English translation (t), the goal is to generate a grounded glossary $G = (k_i, v_i)_{i=1}^m$ where each k_i is a semantically meaningful Sanskrit phrase and v_i is its meaning grounded in the translation (t).

Example Illustration

Sanskrit śloka: नैमिषेऽनिमिषक्षेत्रे ऋषयः शौनकादयः ।
सत्रं स्वर्गायलोकाय सहस्रसममासत ॥

English Translation:

Once, in a holy place in the forest of Naimiṣāranya, great sages headed by the sage Śaunaka assembled to perform a great thousand-year sacrifice for the satisfaction of the Lord and His devotees.

Glossary:

नैमिषे → in the forest known as Naimiṣāranya
अनिमिष-क्षेत्रे → the spot which is especially a favorite of Viṣṇu (who does not close His eyelids) ऋषयः → sages
शौनक-आदयः → headed by the sage Śaunaka
सत्रम् → sacrifice स्वर्गाय → the Lord who is glorified in heaven
लोकाय → and for the devotees who are always in touch with the Lord
सहस्र → one thousand, समम् → years आसत → performed.

Unlike machine translation (Koehn et al., 2003), which targets fluent output over automatically

induced phrase pairs, bilingual lexicon induction (Lample et al., 2018; Artetxe et al., 2017), which derives word-level equivalents from corpus-level statistics, or definition modeling (Noraset et al., 2017), which generates context-free glosses, grounded glossary generation jointly requires recovering morphologically complex phrase boundaries and producing meanings faithful to a specific provided translation. The generated glossary must remain grounded in the provided translation, capturing interpretable lexical units as structured phrase-meaning pairs while avoiding unsupported or hallucinated interpretations.

3 Dataset

3.1 Data Sources

The dataset is curated from two publicly available educational resources: Rāmāyaṇa and Śrīmad Bhāgavatam, and use it for research purposes. Each instance contains:

(Sanskrit śloka, English translation, a glossary mapping Sanskrit phrases)

3.2 Data Preparation

We parse semi-structured glossary annotations into structured key-value dictionaries and apply light normalization, including Unicode and whitespace normalization, while preserving the original linguistic structure of the text.

The dataset is divided into train, validation, and test splits: **Train:** 25,050, **Validation:** 3,133, **Test:** 3,133.

To reduce leakage, contiguous verses and near-duplicate glossary entries are prevented from appearing across splits. Overlapping gloss key between Train, Validation and Test is shown in Table 6.

4 Evaluation Metrics

We evaluate generated glossaries along two complementary dimensions: (i) correctness of generated Sanskrit glossary keys and (ii) semantic faithfulness of generated meanings.

4.1 Jaccard

To measure glossary key recovery, we compute an overlap-based Jaccard score between the set of generated glossary keys (G_p) and reference glossary keys (G_r):

$$\text{Jaccard} = \frac{|G_p \cap G_r|}{|G_p \cup G_r|}$$

where each glossary key corresponds to a Sanskrit phrase segment. This metric evaluates how accurately the model recovers the reference semantic units.

4.2 Meaning Faithfulness

Meaning Faithfulness measures semantic consistency between generated and reference glossary entries. For each generated key, we identify the closest reference key using normalized Levenshtein similarity via RapidFuzz⁵(Bachmann, 2025) and consider it matchable only if the similarity exceeds 70, we mathematically represent this as $\text{KeySim}(k_p, k_r)$. For matched keys, semantic similarity between generated and reference meanings is computed using cosine similarity between SentenceTransformer embeddings from BAAI/bge-base-en-v1.5:

$$\text{ValSim}(v_p, v_r) = \cos(\phi(v_p), \phi(v_r))$$

where $\phi(\cdot)$ denotes the embedding function. Each glossary entry contributes:

$$\text{Contribution} = \text{KeySim}(k_p, k_r) \times \text{ValSim}(v_p, v_r)$$

with $\text{KeySim} \in [0, 1]$, and the final datapoint score is obtained by averaging contributions across all generated entries, and we take average of the this score to evaluate the models. Extra or spurious keys are explicitly penalized: unmatched keys contribute zero, while weakly matched keys receive lower scores due to reduced key and value similarity, thereby lowering the overall average.

Intuition: The intuition behind this design of metric is to explicitly handle three common scenarios in key generation: under segmentation, over segmentation and proper segmentation.

Under Segmentation: In this case, the model fails to split a word that should have been fragmented. Since the ground-truth key is segmented, the maximum Levenshtein similarity between the generated key and the ground truth keys typically decreases. Multiplying this lower KeySim with the cosine similarity therefore penalizes the generated key.

Over segmentation: Here, the model splits a key into fragments more than necessary. As in the previous case, the resulting fragments obtain lower Levenshtein similarity scores with the ground-truth keys, and multiplying by KeySim reduces their contribution, thereby penalizing the model.

In both of the above cases, severe under segmentation or over segmentation may cause the similarity score to fall below the threshold, setting KeySim to 0. (For example, this would happen if the model splits a single word into three fragments or merges three separate words into one.)

Proper segmentation: When the key is segmented correctly, KeySim remains high, 100% for an exact match and close to 100% for near perfect matches. In such cases, the key is effectively not penalized.

5 Methodology

5.1 Models Used

For the task of Glossary generation, we consider the models: gemma-3n-E4B-it (Team, 2025b), gemma-3-12b-it (Team, 2025a), phi-4 (Abdin et al., 2024), Qwen3.5-9B (Qwen Team, 2026). The models translation capability is discussed in Section A.3.

5.2 Experimental Setup

Baseline: For baselines, we use FastAlign(Dyer et al., 2013) as a non-LLM based methodology for generating glossary using the given Sanskrit śloka and it’s English translation. FastAlign is a lightweight statistical word alignment model based on IBM Model 2(Brown et al., 1993) that learns token-level correspondences from parallel bilingual corpora. Additionally, we use ByT5-Sanskrit(Nehrdich et al., 2024) in three configurations: (i) a single Seq2Seq model trained end-to-end to produce the glossary directly from the shloka and translation; (ii) a two-stage pipeline where one ByT5-Sanskrit model is fine-tuned for segmentation (shloka gloss keys) and a second ByT5-Sanskrit model is fine-tuned to generate the glossary from the predicted segments plus the reference translation.

In-Context Learning: We experiment whether the a language model learn to generate the glossary with increased number of examples via zero-shot, few-shot learning with 1, 5 and 10 examples. These examples are available in Section A.8. The system and user prompt used for N-shot generation is available in Section A.6.

Instruction Fine-tuning(IFT): We instruction fine-tune language models using the aforementioned user prompt.

Settings for IFT: We instruction fine-tune all models using LoRA with largely shared hyperparam-

⁵<https://github.com/rapidfuzz/RapidFuzz>

ters, including $\alpha = 16$, dropout 0.05, RS-LoRA, AdamW fused optimization, a learning rate of 3×10^{-5} , cosine scheduling with warmup, BF16 precision, and 10 training epochs. Model-specific variations primarily involved LoRA rank, adapted modules, loss formulation, batch size, and sequence length, with larger models using higher LoRA ranks and gemma-3n-E4B-it employing reduced batch sizes and shorter context lengths due to memory constraints.

6 Result

6.1 Quantitative Analysis

Table 1 reports performance across Jaccard and Meaning Faithfulness respectively.

IFT substantially outperforms both FastAlign and in-context learning across all models and evaluation metrics, with Phi-4 achieving the best overall performance. While increasing the number of in-context examples generally improves results over zero-shot prompting, instruction fine-tuning provides larger and more stable gains, indicating that grounded glossary generation benefits from task-specific adaptation beyond token-level alignment or prompting alone. Additionally, consistently higher Meaning Faithfulness scores than Jaccard scores suggest that models often generate semantically plausible meanings even when phrase segmentation is imperfect. We further compare Jaccard vs Standard Metrics: Precision, Recall and F1-score and analyze the model results on each of these metrics in Section A.5.

6.2 Qualitative Analysis

Error Category	Count
Over-Segmentation	122
General Segmentation Issues	22
Both Over and Under Segmentation	12
Under-Segmentation	9
Semantic & Translation Errors	4
No Issues	5

Table 2: Error analysis findings across 174 generated samples.

We perform qualitative error analysis on 174 low-scoring samples (bottom 5% by Meaning Faithfulness) and categorize errors into six groups (Table 2). The dominant failure mode is *Over-Segmentation* (70%), where Sandhi and Samāsa compounds are incorrectly fragmented,

e.g., trasareṇuḥ(त्रसरेणुः) split into tra(त्र) and sareṇuḥ(सरेणुः).

We also observe concurrent over- and under-segmentation, indicating inconsistent boundary detection, while purely semantic errors are comparatively rare. These results suggest that morphology-aware constraints and improved compound boundary modeling remain critical for Sanskrit glossary generation. A detailed analysis is available in Section A.4.

6.3 Analysis on Seen vs Unseen Gloss Keys

Table 3 reports the correct gloss key predicted results for these two subsets - gloss keys that were seen during training, and gloss keys that were not seen during training. Out of 12,194 seen and 8,956 unseen gloss keys, IFT gains are large on both subsets: for Phi-4, +66 points on seen keys and +60 points on unseen keys, a comparable magnitude. If IFT models were simply memorizing training units, we would expect little to no improvement on unseen keys; instead the gain is nearly as large as on seen keys

7 Ablations

We perform two ablation studies to analyze the role of explicit structural decomposition and robustness to Sanskrit word-order variation on the best performing model, phi-4.

7.1 Ablation 1: Segmentation-Augmented Glossary Generation

Model	Jaccard	Meaning Faithfulness
gemma-3-12b-it	0.656	0.769
phi-4	0.708	0.785
gemma-3n-E4B-it	0.616	0.748
Qwen3.5-9B	0.696	0.783
phi-4 + gold segmentations	1.0	0.872
phi-4 + ByT5-Sanskrit predicted segments	0.501	0.741

Table 4: Performance of segmentation-augmented glossary generation with instruction-tuned models that first segment the 'sloka and then generate meanings.

In this ablation, we instruction fine-tune models to first generate semantic Sanskrit segments and then produce grounded glossary meanings, in order to evaluate the impact of explicit intermediate segmentation. Table 4 results show that although performance is nearly identical to direct glossary

Model	Jaccard					Meaning Faithfulness				
	IFT	0 shot	1 shot	5 shot	10 shot	IFT	0 shot	1 shot	5 shot	10 shot
FastAlign	0.144		—			0.32		—		
ByT5-Sanskrit (Seq2Seq)	0.543		—			0.550		—		
ByT5-Sanskrit Pipeline (Segment $\Rightarrow$ Glossary)	0.520		—			0.552		—		
gemma-3-12b-it	0.666	0.238	0.278	0.304	0.314	0.768	0.517	0.557	0.546	0.566
phi-4	0.716	0.092	0.358	0.230	0.299	0.787	0.182	0.635	0.604	0.628
gemma-3n-E4B-it	0.617	0.175	0.257	0.285	0.281	0.746	0.473	0.557	0.572	0.565
Qwen3.5-9B	<u>0.693</u>	0.296	0.377	0.376	0.378	<u>0.781</u>	0.570	0.631	0.628	0.636

Table 1: **Jaccard** measures key correctness, while **Meaning Faithfulness** measures the semantic accuracy of values against the ground truth. Both are averaged per entry.

Model	Seen - Zero-shot	Seen - IFT	Unseen - Zero-shot	Unseen - IFT
Phi-4	2626 (21.53%)	10713 (87.85%)	564 (6.30%)	5952 (66.45%)
Qwen3.5-9B	6263 (51.36%)	10524 (86.30%)	2438 (27.22%)	5809 (64.86%)
Gemma-3-12B-it	<u>5481 (44.94%)</u>	10413 (85.40%)	<u>1952 (21.79%)</u>	5400 (60.29%)
Gemma-3n-E4B-it	3738 (30.65%)	9870 (80.94%)	1606 (17.93%)	4924 (54.98%)
ByT5-Sanskrit (Seq2Seq)	—	5,284 (43.33%)	—	2,529 (28.24%)
ByT5-Sanskrit Pipeline (Segment $\Rightarrow$ Glossary)	—	8,375 (68.68%)	—	4,414 (49.29%)

Table 3: Correct gloss key predicted - gloss keys seen vs not seen during training.

generation for the models selected but it is seen that by providing the gold segmentations to generate only the glossary, we found that the Meaning Faithfulness score increased from 0.787 to 0.872. This indicates that segmentation quality plays an important role not only in recovering the correct key set, but also in improving the semantic faithfulness of the generated glossary.

7.2 Ablation 2: Robustness to Word Order Perturbation

Sanskrit permits relatively free word order while preserving semantic relationships through morphology and syntax. To evaluate robustness to word-order variation, we randomly shuffle words within each śloka line while preserving the original line boundaries delimited by “I” and “II”. [Table 5](#) vs [Table 1](#), shows that word-order perturbation causes noticeable degradation in performance, particularly for Qwen3.5-9B, indicating that current models still rely substantially on sequential structure despite Sanskrit’s relatively free word order.

The pre-trained models may have seen publicly available Sanskrit texts or analyses, but their poor zero-shot performance suggests they are not simply reproducing memorized material. The large gains from instruction fine-tuning show that this task needs task-specific adaptation. The word-

order perturbation results further indicate that, even after adaptation, the models rely on sequential cues and are not fully robust to Sanskrit’s relatively free word order.

Model	Jaccard	Meaning Faithfulness
gemma-3-12b-it	<u>0.587</u>	<u>0.750</u>
phi-4	0.632	0.768
Qwen3.5-9B	0.394	0.682

Table 5: Performance under word-order perturbation: words within each śloka line are randomly shuffled.

8 Conclusion

We introduce grounded glossary generation as a structured task requiring recovery of semantically meaningful Sanskrit phrases and translation-grounded glossary meanings. We further present a benchmark dataset, evaluation framework, and systematic study of large language models, showing that instruction fine-tuning substantially improves glossary generation while explicit segmentation supervision affect and word-ordering does affect the performance.

Our results identify morphology and compound boundary detection as the primary bottlenecks for Sanskrit glossary generation. Future work can explore morphology-aware modeling, retrieval-grounded generation, and multilingual lexico-

graphic systems for low-resource settings.

Limitations

- Our dataset is drawn exclusively from two Indian canonical texts - the Vālmīki Rāmāyaṇa and Śrīmad Bhāgavatam - which may limit generalization to other Sanskrit literary traditions with distinct compositional styles, such as Kāvya or Vedic prose.
- The evaluation metrics, Jaccard and Meaning Faithfulness, rely on surface-level string matching and embedding similarity respectively; neither accounts for morphological equivalence between valid sandhi resolutions, potentially penalizing correct but orthographically distinct segmentations.
- Instruction fine-tuning was conducted under memory-constrained settings, necessitating reduced batch sizes and shorter sequence lengths, which may underestimate its performance ceiling. All models were evaluated with LoRA-based adaptation rather than full fine-tuning, leaving open the question of whether additional parameters would further close the segmentation gap.
- Finally, our error analysis is restricted to the bottom 5% of samples by Meaning Faithfulness score, which may not be representative of failure modes across the full score distribution.

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A Appendix

A.1 Dataset Gloss Key Overlap between Train, Validation and Test

Split	Unique Gloss Keys	Seen in Train	Percentage
Train	101,573	-	-
Validation	21,048	12,073	57.36%
Test	21,150	12,194	57.65%

Table 6: Dataset Gloss Key Overlap Information

A.2 Hyperparameters used for LLM

All models were fine-tuned using LoRA with largely shared hyperparameters. Common settings included LoRA scaling ($\alpha = 16$), dropout (0.05), RS-LoRA, AdamW fused optimization, a learning rate of 3×10^{-5} , cosine scheduling with 0.15 warmup, BF16 precision, gradient clipping (1.0), and training for 10 epochs. Checkpointing and evaluation were performed periodically, with the best model selected using validation loss.

Model-specific differences mainly involved LoRA rank, target modules, batch size, and sequence length. gemma-3-12b-it used $r = 128$ with adaptation on all linear layers, while phi-4 and Qwen3.5-9B used $r = 256$. Qwen3.5-9B employed assistant_only_loss, whereas the others used completion_only_loss. gemma-3n-E4B-it applied LoRA only to attention and feed-forward projection layers, used smaller batch sizes (2×4) for per-device batch size and gradient accumulation. Due to memory constraints, we disabled gradient checkpointing, and used shorter input sequences (512 tokens) compared to gemma-3-12b-it where we used (4×4) while phi-4 and Qwen3.5-9B used (8×2).

A.3 Evaluating selected models Translation capability

According to Table 7, the translation-quality ranking tracks the Jaccard ranking: Phi-4 leads both; Qwen, Gemma-3, and Gemma-3n cluster in the middle. This confirms the hypothesis: *how well the model was exposed to Sanskrit data in its pre-training and to its ability to understand the Sanskrit-to-English translation task shedding further light on the relative strengths of these models*, rather than challenging our contribution i.e. Grounded Glossary Generation performance connects to pre-training Sanskrit exposure, a useful diagnostic property of the benchmark. It does not alter our core claim that segmentation is the dominant bottleneck, which the ground-truth-substitution experiment isolates independently of translation ability.

Model	BLEU	ChrF	Semantic Similarity
Phi-4	6.21	30.61	73.77%
Qwen3.5-9B	4.96	28.11	70.20%
Gemma-3n-E4B-it	4.93	27.70	70.57%
Gemma-3-12B-it	4.69	27.31	70.83%

Table 7: Comparison of model performance across BLEU, ChrF, and semantic similarity.

A.4 Qualitative Analysis of Phi-4

To better understand model failures beyond quantitative metrics, we conducted a qualitative error analysis on 174 representative samples (bottom 5% by Meaning Faithfulness score). We grouped the errors into six mutually exclusive categories based on mismatches between the reference and generated glossary entries (Table 2).

The most frequent issue was *Over-Segmentation* (70% of cases), where compounds formed through Sandhi and Samasa were fragmented into smaller units, leading to loss of contextual meaning. For example, trasareṇuḥ(त्रसरेणुः) was incorrectly split into tra(त्र) and sareṇuḥ(सरेणुः).

We also observed cases of *Concurrent Over and Under Segmentation*, where the model simultaneously failed to split some compounds while over-splitting adjacent phrases, indicating inconsistent boundary detection in longer Sanskrit strings. *Under-Segmentation* was less common but still harmful, as in *puruṣa-paricaryay=a*(पुरुष-परिचर्याया), which remained unsplit despite the reference expecting finer segmentation.

A smaller subset of errors were purely *Semantic*, where the generated gloss was contextually incorrect, such as translating *ambaṣṭha*(अम्बष्ठ) as ‘driver’ instead of ‘elephant-keeper.’

For the 5 no issues samples, there were no issues on the generated output but rather the key:val pairs were wrongly translated in the ground truth glossary.

Overall, these findings highlight the need for stronger morphological constraints and improved boundary detection to reduce over-fragmentation in Sanskrit glossary generation.

A.5 Comparing Jaccard against Precision, Recall, F1-score

Precision, Recall, are also valid alternatives for evaluating glossary-unit recovery. We chose Jaccard because the task is formulated as recovering the set of glossary units, and Jaccard provides a simple set-overlap measure that jointly penalizes missing and spurious units. Since Jaccard and F1 are closely related monotonic measures, we do not expect the relative ranking of models to change substantially which is also observed in Table 8. Similarly for Ablation 1, the Jaccard vs Precision, Recall and F1-score, the respective scores are available in Table 9.

Model	Precision	Recall	F1-score	Jaccard
Phi-4	0.8145	0.8237	0.8175	0.716
Qwen3.5-9B	0.8016	0.8041	0.8012	0.693
Gemma-3-12B-it	0.7775	0.7875	0.7807	0.666
Gemma-3n-E4B-it	0.7439	0.7453	0.7426	0.617

Table 8: The metric values for our test data, for correct gloss key identification

Model	Precision	Recall	F1-score	Jaccard
Phi-4	0.8102	0.8173	0.8120	0.708
Qwen3.5-9B	0.8055	0.8042	0.8031	0.696
Gemma-3-12B-it	0.7770	0.7732	0.7732	0.656
Gemma-3n-E4B-it	0.745	0.742	0.743	0.616

Table 9: Ablation 1 metric values for our test data, for correct gloss key identification

A.6 Prompts used for N-shot and IFT

System Prompt

You are a Sanskrit-to-English glossary extraction engine. Given a Sanskrit shloka and its English translation, output a single JSON object mapping each Sanskrit word to its English meaning.

Rules:

1. Output ONLY a Python dict. Start with "{", end with "}". No other text.
2. Keys are Sanskrit words (or sandhi-resolved tokens) as they appear in the shloka.
3. Values are their English meanings as evidenced by the translation.
4. No markdown, no code fences, no explanation, no preamble.

Format: {"glossary": {"sanskrit_word": "english_meaning", ...}}

User Prompt

Generate a Sanskrit-English glossary mapping for the following shloka based on its translation.
Shloka: {shloka}
Translation: {translation}

A.7 Prompts used for Ablation 1

System Prompt for Ablation 1

Perform Padaccheda (resolve sandhi, keep samasa compound words intact). Analyze the shloka word by word using the translation as a reference. End your reasoning with the resolved word sequence as: [word1, word2, ...].

STEP 2 FINAL GENERATION:

Output the glossary as a code block. Rules for the output:

1. Output exactly ONE valid Python dict with a single key 'glossary'.
2. All keys are the Padaccheda-resolved Sanskrit words, in the order they appear in the shloka.
3. All values are their English meanings, derived strictly from the provided translation.
4. No trailing commas.
5. No extra keys, no nested objects flat dictionary only.

RESPONSE FORMAT (follow exactly, do not deviate):

REASONING:

<your padaccheda analysis ending with the resolved word list>

FINAL GENERATION:

{“glossary”: {“sanskrit_word”: “english_meaning”, ...}}

Do NOT include any text outside these two sections. Do NOT add greetings, notes, or explanations after the output block.

User Prompt for Ablation 1 IFT

INPUT: You are an expert in Sanskrit. Perform Padaccheda (word resolution) on the following Sanskrit shloka, then generate the glossary mapping using the generated Padaccheda and provided English translation.

Shloka: <shloka>

Translation: <translation>

A.8 N-shot Examples

The following data was used as examples for all N-shot experiments. For 1-shot, the first example, for 5-shot, the first 5 examples and similarly for 10-shot all the 10 examples were used.

Shlok: न वयं क्लेशबीजानि यतः स्युः पुरुषर्षभ । पुरुषं तं विजानीमो वाक्यभेदविमोहिताः ।

Translation: O greatest among human beings, it is very difficult to ascertain the particular miscreant who has caused our sufferings, because we are bewildered by all the different opinions of theoretical philosophers.

Glossary: {'न': 'not', 'वयम्': 'we', 'क्लेश-बीजानि': 'the root cause of sufferings', 'यतः': 'wherefrom', 'स्युः': 'it so happens', 'पुरुष-ऋषभ': 'O greatest of all human beings', 'पुरुषम्': 'the person', 'तम्': 'that', 'विजानीमः': 'know', 'वाक्य-भेद': 'difference of opinion', 'विमोहिताः': 'bewildered by'}

Shlok: सोऽधिक्षिप्तो दुर्वाचोभिः पदाहत इवोरगः । निश्चक्राम गदापाणिरमर्षात्ताम्रलोचनः ।

Translation: Offended by these harsh words, śambara became as agitated as a kicked snake. He came out, club in hand, his eyes red with rage.

Glossary: {'सः': 'he', 'अधिक्षिप्तः': 'insulted', 'दुर्वाचोभिः': 'by harsh words', 'पदा': 'by a foot', 'आहतः': 'struck', 'इव': 'like', 'उरगः': 'a snake', 'निश्चक्राम': 'came out', 'गदा': 'a club', 'पाणिः': 'in his hand', 'अमर्षात्': 'out of intolerant anger', 'ताम्र': 'copper-red', 'लोचनः': 'whose eyes'}

Shlok: दृष्ट्वान्यांश्च महोत्पातानतत्तत्त्वविदः प्रजाः । ब्रह्मपुत्रानृते भीता मेनिरे विश्वसम्प्लवम् ।

Translation: Marking these and many other omens of evil times, everyone but the four sage sons of Brahmā, who were aware of the fall of Jaya and Vijaya and of their birth as Diti's sons, was seized with fear. They did not know the secrets of these portents and thought that the dissolution of the universe was at hand.

Glossary: { 'दृष्ट्वा': 'having seen', 'अन्यान्': 'others', 'च': 'and', 'महा': 'great', 'उत्पातान्': 'evil omens', 'अ-तत्-तत्त्व-विदः': 'not knowing the secret (of the portents)', 'प्रजाः': 'people', 'ब्रह्म-पुत्रान्': 'the sons of Brahmā (the four Kumāras)', 'ऋते': 'except', 'भीताः': 'being fearful', 'मेनिरे': 'thought', 'विश्व-सम्प्लवम्': 'the dissolution of the universe' }

Shlok: कथं वर्तेत विहरेत् कैर्वा ज्ञायेत लक्षणैः । किं भुञ्जीतोत विसृजेच्छयीतासीत याति वा । एतदच्युत मे ब्रूहि प्रश्नं प्रश्नविदां वर । नित्यबद्धो नित्यमुक्त एक एवेति मे भ्रमः ।

Translation: O my Lord, Acyuta, the same living entity is sometimes described as eternally conditioned and at other times as eternally liberated. I am not able to understand, therefore, the actual situation of the living entity. You, my Lord, are the best of those who are expert in answering philosophical questions. Please explain to me the symptoms by which one can tell the difference between a living entity who is eternally liberated and one who is eternally conditioned. In what various ways would they remain situated, enjoy life, eat, evacuate, lie down, sit or move about?

Glossary: { 'कथम्': 'in what way', 'वर्तेत': 'he is situated', 'विहरेत्': 'he enjoys', 'कैः': 'by which', 'वा': 'or', 'ज्ञायेत': 'would be known', 'लक्षणैः': 'by symptoms', 'किम्': 'what', 'भुञ्जीत': 'he would eat', 'उत': 'and', 'विसृजेत्': 'would evacuate', 'शयीत': 'would lie down', 'आसीत': 'would sit', 'याति': 'goes', 'एतत्': 'this', 'अच्युत': 'O Acyuta', 'मे': 'to me', 'ब्रूहि': 'explain', 'प्रश्नम्': 'the question', 'प्रश्न-विदाम्': 'of all those who know how to answer questions', 'वर': 'O the best', 'नित्य-बद्धः': 'eternally conditioned', 'नित्य-मुक्तः': 'eternally liberated', 'एकः': 'singular', 'एव': 'certainly', 'इति': 'thus', 'भ्रमः': 'confusion' }

Shlok: तत्रेतिकृत्यमुपशिक्ष यथोपदेशं येनैष मे कर्षितोऽतिरिंसयात्मा । सिद्ध्येत ते कृतमनोभवधर्षिताया दीनस्तदीश भवनं सदृशं विचक्ष्व ।

Translation: Devahūti continued: My dear lord, I am struck by excited emotion for you. Therefore kindly make what arrangements must be made according to the scriptures so that my skinny body, emaciated through unsatisfied passion, may be rendered fit for you. Also, my lord, please think of a suitable house for this purpose.

Glossary: { 'तत्र': 'in that', 'इति-कृत्यम्': 'what is necessary to be done', 'उपशिक्ष': 'perform', 'यथा': 'according to', 'उपदेशम्': 'instruction in scripture', 'येन': 'by which', 'एष': 'this', 'मे': 'my', 'कर्षितः': 'emaciated', 'अतिरिंस-सया': 'due to intense passion not being satisfied', 'आत्मा': 'body', 'सिद्ध्येत': 'it may be rendered fit', 'ते': 'for you', 'कृत': 'excited', 'मनः-भव': 'by emotion', 'धर्षितायाः': 'who am struck', 'दीनः': 'poor', 'तत्': 'therefore', 'ईश': 'O my dear lord', 'भवनम्': 'house', 'सदृशम्': 'suitable', 'विचक्ष्व': 'please think of' }

Shlok: न वेद धर्मं किल पद्मयोनि- न ब्रह्मपुत्रा भृगुनारदाद्याः । न वै कुमारः कपिलो मनुश्च ये नो निषेधन्त्यतिवर्तिनं हरम् ।

Translation: Alas, Lord Brahmā, who has taken his birth from the lotus flower, does not know the principles of religion, nor do the great saints like Bhṛgu and Nārada, nor the four Kumāras, headed by Sanat-kumāra. Manu and Kapila have also forgotten the religious principles. I suppose it to be because of this that they have not tried to stop Lord Śiva from behaving improperly.

Glossary: { 'न': 'not', 'वेद': 'knows', 'धर्मम्': 'the religious principles', 'किल': 'indeed', 'पद्म-योनिः': 'Lord Brahmā', 'ब्रह्म-पुत्राः': 'the sons of Lord Brahmā', 'भृगु': 'Bhṛgu', 'नारदः': 'Nārada', 'आद्याः': 'and so on', 'वै': 'indeed', 'कुमारः': 'the four Kumāras (Sanaka)', 'कपिलः': 'Lord Kapila', 'मनुः': 'Manu himself', 'च': 'and', 'ये': 'who', 'नो': 'not', 'निषेधन्ति': 'order to stop', 'अति-वर्तिनम्': 'who is beyond laws and orders', 'हरम्': 'Lord Śiva' }

Shlok: कर्माण्यारभते येन पुमानिह विहाय तम् । अमुत्रान्येन देहेन जुष्टानि स यदश्नुते ।

Translation: The results of whatever a living entity does in this life are enjoyed in the next life.

Glossary: { 'ऽकर्माणि': 'fruitive activities', 'आरभते': 'begins to perform', 'येन': 'by which', 'पुमान्': 'a living entity', 'इह': 'in this life', 'विहाय': 'giving up', 'तम्': 'that', 'अमुत्र': 'in the next life', 'अन्येन': 'another', 'देहेन': 'by a body', 'जुष्टानि': 'the results', 'सः': 'he', 'यत्': 'that', 'अश्नुते': 'enjoys' }

Shlok: आरोप्यारुरुहे यानं भ्रातृभ्यां हनुमद्युतः । विभीषणाय भगवान् दत्त्वा रक्षोगणेशताम् । लङ्कामायुश्च कल्पान्तं ययौ चीर्णव्रतः पुरीम् ।

Translation: After giving Vibhīa the power to rule the Rākasa population of Lāṅkā for the duration of one kalpa, Lord Rāmacandra, the Supreme Personality of Godhead [Bhagavān], placed Sītādevī on an airplane decorated with flowers and then got on the plane Himself. The period for His living in the forest having ended, the Lord returned to Ayodhyā, accompanied by Hanumān, Sugrīva and His brother Lakmaā.

Glossary: { 'आरोप्य': 'keeping or placing', 'आरुरुहे': 'got up', 'यानम्': 'on the airplane', 'भ्रातृभ्याम्': 'with His brother Lakmaā and the commander Sugrīva', 'हनुमत्-युतः': 'accompanied by Hanumān', 'विभीषणाय': 'unto Vibhīa', 'भगवान्': 'the Lord', 'दत्त्वा': 'gave charge', 'रक्ष-गण-ईशताम्': 'the power to rule over the Rākasa population of Lāṅkā', 'लङ्काम्': 'the state of Lāṅkā', 'आयुः च': 'and the duration of life', 'कल्प-अन्तम्': 'for many', 'ययौ': 'returned home', 'चीर्ण-व्रतः': 'finishing the duration of time living in the forest', 'पुरीम्': 'to Ayodhyā-purī' }

Shlok: एतां विद्यामधिगतो विश्वरूपाच्छतक्रतुः । त्रैलोक्यलक्ष्मीं बुभुजे विनिर्जित्य मृधेऽसुरान् ।

Translation: King Indra, who performed one hundred sacrifices, received this prayer of protection from Viśvarūpa. After conquering the demons, he enjoyed all the opulences of the three worlds.

Glossary: { 'एताम्': 'this', 'विद्याम्': 'prayer', 'अधिगतः': 'received', 'विश्वरूपात्': 'from the brāhmaa Viśvarūpa', 'शत-क्रतुः': 'Indra', 'त्रैलोक्य-लक्ष्मीम्': 'all the opulence of the three worlds', 'बुभुजे': 'enjoyed', 'विनिर्जित्य': 'conquering', 'मृधे': 'in battle', 'असुरान्': 'all the demons' }

Shlok: ननृतुस्तस्य पुरतः स्त्रियोऽथो गायका जगुः । मृदङ्गवीणापणवैर्वाद्यं चक्रुर्मनोरमम् ।

Translation: The women danced before the sage, and the celestial singers sang to the charming accompaniment of drums, cymbals and vīās.

Glossary: { 'ननृतुः': 'danced', 'तस्य': 'of him', 'पुरतः': 'in front', 'स्त्रियः': 'women', 'अथ उ': 'and furthermore', 'गायकाः': 'singers', 'जगुः': 'sang', 'मृदङ्ग': 'with drums', 'वीणा': 'stringed instruments', 'पणवैः': 'and cymbals', 'वाद्यम्': 'instrumental music', 'चक्रुः': 'they made', 'मनः-रमम्': 'charming' }